FORECASTING CAPITAL INVESTMENTS AND CAPITAL STRUCTURE

Learning Outcome

- explain approaches to forecasting a company's capital investments and capital structure

Projections for long-term assets are based on cash flow statement and income statement projections, because net PP&E and intangible assets on the balance sheet primarily increase due to capital expenditures and decrease due to depreciation and amortization expenses. Capital expenditures can be broken down into maintenance capital expenditures necessary to sustain the current business and growth capital expenditures needed to expand the business. Maintenance capital expenditure forecasts are often based on historical depreciation and amortization expenses, usually with a small adjustment upward to account for inflation in capital goods. For businesses with low fixed asset turnover, maintenance capital expenditure requirements can be quite high. Growth capital expenditure forecasts are more discretionary and are tied to management's expansion plans and revenue growth. Depreciation and amortization forecasts are based on net PP&E and intangibles on the balance sheet (which increase due to capital expenditures) and their useful lives as assumed by management's accounting policies, which can be approximated by the ratio of gross fixed assets to depreciation and amortization expenses. Information may also be found in the notes to the financial statements.

Analysts must also make projections about a company's future capital structure. Leverage ratios—such as debt to capital, debt to equity, and debt to EBITDA—are often used as the forecast object to project future debt and equity levels. Analysts should consider historical company practice, management's financial strategy, and the capital requirements implied by the capital expenditure assumptions when projecting the future capital structure. Management may provide guidance on target capital structure, debt covenant ratios (e.g., net debt to EBITDA), and capital expenditures, sometimes broken down into maintenance, growth, and acquisitions. Example 3 demonstrates the use of these forecast objects for YY Ltd., the same company in Example 2.

EXAMPLE 3

Balance Sheet Modeling

Exhibit 10 shows financial data for YY Ltd. related to its PP&E and intangible assets. Based on the data in Exhibit 10 and the data and analysis in Example 2, answer Questions 1 and 2.

Exhibit 10:

YY Ltd. Long-Term Asset Data (CNY millions)

Year	1	2	3
PP&E, net	5,068	6,992	6,306
Goodwill	282	248	253
Intangible assets, net	1,779	1,424	4,013
Total fixed assets	7,129	8,664	10,572

Capital expenditures − intangibles	333	142	3,310
Depreciation expense	220	324	518
Amortization expense	529	486	666

Note: PP&E and intangibles asset account balances were also affected each year by changes in exchange rates and by disposals, which are not shown in the exhibit. Assume that such effects are zero in years 4–6.

1. Calculate the following:

A. Capital expenditures (for both PP&E and intangibles) as a percentage of revenue (for years 1–3)

B. Depreciation expense as a percentage of beginning-of-year PP&E, net (for years 2 and 3)

C. Amortization expense as a percentage of beginning-of-year intangible assets, net (for years 2 and 3).

Hide Solution

Solution:

Using the data, the following percentages were calculated.

Year	1	2	3
Revenue	174,915	205,839	245,866
Capital expenditures − PP&E (% of revenue)	2.2%	1.7%	1.2%
Capital expenditures − intangibles (% of revenue)	0.2%	0.1%	1.3%
Depreciation (% of beginning PP&E)		6%	7%
Amortization (% of beginning intangibles)		27%	47%

2. Given the following assumptions and forecast revenue from Example 2, calculate expected total fixed assets for years 4–6.

- Capital expenditures for PP&E as a percentage of revenue to remain at the year 3 level
- Capital expenditures for intangibles as a percentage of revenue to remain at the year 1 level
- Goodwill to remain at the year 3 level
- Depreciation and amortization expenses as a percentage of beginning-of-year PP&E, net, and intangible assets, net, to remain at year 3 levels

Hide Solution

Solution:

Using the data provided, expected total fixed assets for years 4–6 was calculated as CNY12,351 million; CNY15,179 million; and CNY18,662 million, respectively. PP&E, net, for each year was calculated as the prior period balance plus capital expenditures minus depreciation expense. Intangible assets, net, for each year was calculated in a similar fashion. Goodwill was held constant at CNY253 million.

Year	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
PP&E, net	5,068	6,992	6,306	9,410	12,854	16,583
Goodwill	282	248	253	253	253	253
Other intangible assets, net	1,779	1,424	4,013	2,688	2,072	1,827

assets	7,129	8,664	10,572	12,351	15,179	18,662
Capital expenditures – PP&E	3,785	3,405	3,026	3,571	4,142	4,680
Capital expenditures – PP&E (% of revenue)	2.2%	1.7%	1.2%	1.2%	1.2%	1.2%
Capital expenditures – intangibles	333	142	3,310	552	641	724
Capital expenditures – intangibles (% of revenue)	0.2%	0.1%	1.3%	0.2%	0.2%	0.2%
Depreciation expense	220	324	518	467	697	952
Depreciation (% of beginning PP&E)		6%	7%	7%	7%	7%
Amortization expense	529	486	666	1,877	1,257	969
Amortization (% of beginning intangibles)		27%	47%	47%	47%	47%

3. Based on Exhibit 11, which shows financial data for YY Ltd. related to its capital structure and profitability, answer Question 3.

Exhibit 11:

YY Ltd. Debt and Profitability Data (CNY millions)

Year	1	2	3
Gross debt	10,931	17,624	17,597
Revenue	174,915	205,839	245,866
EBITDA	9,304	12,343	14,190
EBITDA margin	5.3%	6.0%	5.8%

YY Ltd. management has a year 6 gross debt to EBITDA ratio target of 2.0.

A. Assuming an EBITDA margin of 6.0%, revenue forecasts from Example 2, and gross debt to EBITDA ratios of 1.25, 1.50, and 2.0 for years 4, 5, and 6, respectively, calculate expected gross debt for years 4–6.

B. Given the results of part A, how much incremental borrowing does the forecast imply from year 3 to year 6?

Hide Solution

Solution:

Gross debt of CNY21,579 million; CNY30,289 million; and CNY45,635 million are estimated for years 4–6, respectively, for YY Ltd. This forecast is made by first multiplying forecast revenue by the forecast EBITDA margin to calculate forecast EBITDA. Then, the expected gross debt to EBITDA ratio is multiplied by the forecast EBITDA to calculate forecast gross debt.

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Revenue	174,915	205,839	245,866	290,122	336,541	380,292
EBITDA	9,304	12,343	14,190	17,407	20,192	22,818
EBITDA margin	5.3%	6.0%	5.8%	6.0%	6.0%	6.0%
Gross debt to EBITDA	1.17	1.43	1.24	1.25	1.50	2.00

Exhibit 12 discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s capital expenditures and gross debt to EBITDA.

Exhibit 12:

Forecast Approaches for Warehouse Club Inc.'s Capital Expenditures and Structure

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Capital expenditures as a percentage of net sales	Historical results	Warehouse has spent 2% of net sales on capital expenditures for the last 10 years, which Nguyen can use as the forecast going forward.
	Historical base rates and convergence	More mature retailers (e.g., Walmart in the USA) have all but stopped opening new stores, which has reduced their capital requirements to maintenance and new initiatives like e-commerce. If Warehouse does not intend to go to new geographies or launch an e-commerce business, Nguyen might forecast capital expenditures trending down to maintenance level, along with a decline in new-store openings per year, as Warehouse matures.
	Management guidance	Warehouse management provides guidance on new-store openings, which Nguyen can use to forecast capital expenditures (along with a maintenance capital expenditure forecast and a forecast for growth capital expenditures per new store).
	Analyst's discretionary forecast	Nguyen might instead use a discretionary forecast for capital expenditures by projecting a maintenance CAPEX amount (not disclosed by management) and growth CAPEX associated with new-store openings.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross debt to EBITDA	Historical results	For the last five years, Warehouse's ratio of debt to trailing twelve months EBITDA has been 1.10, which Nguyen can project going forward. The product of this ratio and projected EBITDA (from earlier forecasts of sales and operating expenses) is Warehouse's gross debt.
	Historical base rates and convergence	Warehouse's more mature competitors and peers use a greater amount of debt in their capital structure, and return more capital to shareholders with buybacks and dividends. Nguyen might forecast the same for Warehouse over time, with an increasing ratio of debt to EBITDA and percentage of free cash flow returned to shareholders.

Management guidance	Warehouse's management may provide a target capital structure through a target debt to EBITDA ratio or credit rating, which Nguyen can use as the forecast.
Analyst's discretionary forecast	Nguyen might instead make a discretionary forecast linked to her free cash flow forecast, projecting a debt to EBITDA ratio and return of capital to shareholders. Depending on Fyleton Investments' strategy, Nguyen might engage with Warehouse's management to advocate for such a capital structure.

Question Set

An analyst gathers the following information related to Iliso's capital expenditures and capital structure (USD millions):

	2X15	2X16	2X17	2X18	2X19
Iliso's retail sales	2,589	3,424	5,100	6,052	7,336
Operating income	−1	18	61	72	130
Depreciation expense	25	43	65	91	123
PP&E (net)	198	294	435	590	768
Short-term debt	56	255	198	66	0
Long-term debt	43	157	168	144	144
Equity	548	732	858	1,006	1,178
Capital expenditures	79	140	208	247	301

1. Which of the following would the analyst *most likely* use in estimating Iliso's maintenance capital expenditures?

A. Operating income
B. Depreciation expense
C. Earnings before interest, taxes, depreciation, and amortization (EBITDA)

Hide Solution

Solution:

B is correct. Depreciation expense can serve as the basis for maintenance capital expenditures, as it is management's estimate of the cost of fixed assets expensed on the income statement in a manner that tracks its use.

2. Which of the following approaches is *most* appropriate for forecasting Iliso's capital expenditures?

A. Historical results
B. Analyst's discretionary forecast
C. Historical base rates and convergence

Hide Solution

Solution:

A is correct. Capital expenditures have been 4.1% of retail sales for the past four years (2X16–2X19). The historical results approach—using the bottom-up driver retail sales—is most appropriate. An analyst's discretionary forecast is not required, because Iliso is not in a cyclical industry or undergoing a fundamental change. Iliso's unique model and the breadth of the industry make the approach using historical base rates and convergence to the industry inappropriate.

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expense of 20% of opening PP&E (net), and debt/EBITDA of 0.50. The analyst's forecast for debt (in millions) is *closest* to:

A. 84.
B. 161.
C. 195.

Hide Solution

Solution:

B is correct.

2X20 operating income = 0.02(8,411) = 168

2X20 depreciation expense = 0.20(768) = 154

EBITDA = 168 + 154 = 322

Debt = 0.50(322) = 161

4. The analyst notes that Iliso's common stock and paid-in capital accounts increased, in total, by 75 million from 2X18 to 2X19, but only 42 million was reported as stock issuance on the statement of cash flows. Retained earnings increased by the amount of net income. Which of the following *most likely* explains the difference of 33 million?

A. A stock split
B. A stock dividend
C. Stock-based compensation

Hide Solution

Solution:

C is correct. Stock-based compensation is a non-cash expense in the income statement and is reported as an addition to net income in the indirect method of reporting cash flow from operations. The issuance also increases common stock and paid-in capital. A stock split typically simply increases the number of shares outstanding and decreases any stated par value per share. A stock dividend increases common stock and paid-in capital and reduces retained earnings by the same amount.

Discuss Filter

Discussion

What do you want to discuss... (required)

MO

Example 3 Q2. The question asks us to calculate expected total fixed assets for years 4–6, however does not provide the growth rate factors for PP&E, net and Intangible assets, net.

What are the assumptions for PP&E, net growth factor as well as Other intangible assets, net growth factor?

For example, if we take PP&E as a percentage of revenue, the percentages are inconsistent, at 3.24%, 3.82%, 4.36%, then if we take year over year growth of PP&E net itself from the previous year, the percentages are also inconsistent at 49.22%, 36.60%, and 29.01%.

There is zero explanation as to how they've arrived at these figures:

PP&E Net Year 4 = 9,410

PP&E Net Year 5 = 12,854

PP&E Net Year 6 = 16,583

Other intangible assets, net Year 4 = 2,688

Other intangible assets, net Year 5 = 2,072

Other intangible assets, net Year 6 = 1,827

What am I missing?

Created 4 months ago by Maxim Ovchinnikov

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